CASE STUDY REPORT

Lab: Internal web server

In progress Created October 19, 2026 Last updated October 19, 2026

6 DEVICES	**1** COMPROMISED	**6** OPEN PORTS
12 MAP CHANGES	**3** ACTIONS	**2** NOTES

TARGET

10.10.20.0/24 (practice lab)

DESCRIPTION

Not filled in

SCOPE

Hosts in 10.10.20.0/24 only. No denial of service.

OBJECTIVE

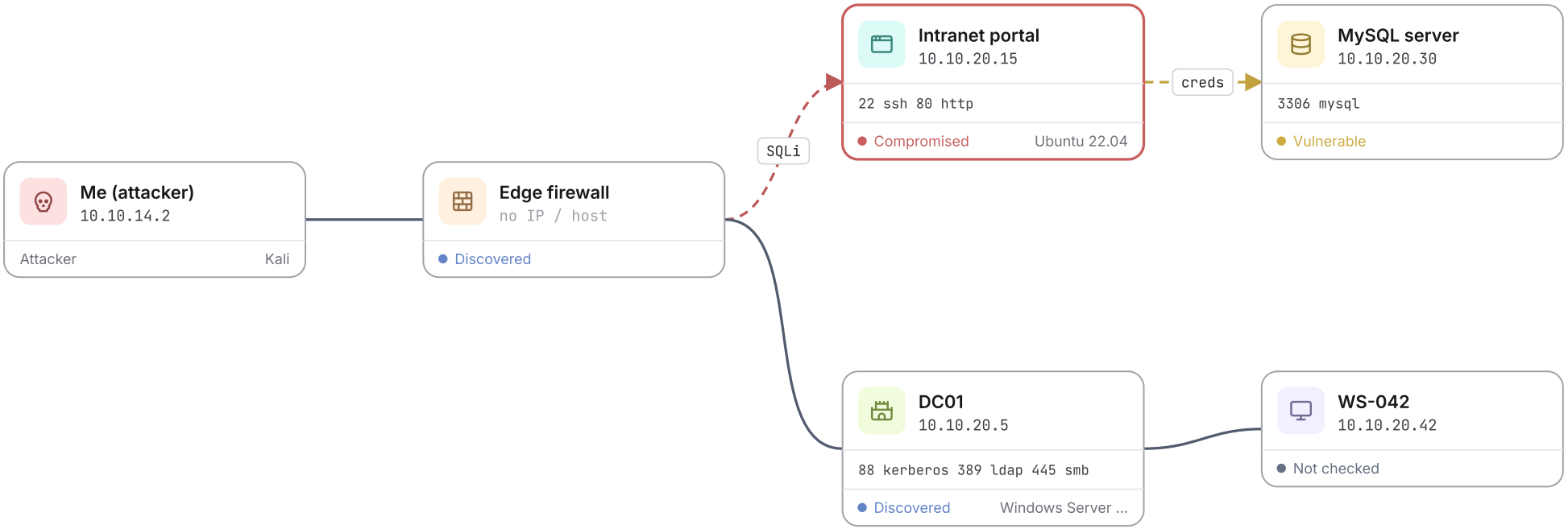
Get a foothold, then reach the database and document the path.

STARTING INFORMATION

VPN access to the lab network. No credentials given.

Network map

Lab: Internal web server



Not checked · Discovered · Vulnerable · Compromised · Out of scope

Network link · Attack path · Pivot / tunnel · Data flow

Device details

6 devices

Me (attacker) Attacker `10.10.14.2` Kali

No ports recorded.

Edge firewall Firewall — Discovered

No ports recorded.

Intranet portal Web app `10.10.20.15` Ubuntu 22.04 — Compromised

PORT	STATE	SERVICE	VERSION	NOTE
22/tcp	open	ssh	OpenSSH 8.9	
80/tcp	open	http	nginx 1.18	login form, SQLi in username

MySQL server Database `10.10.20.30` — Vulnerable

PORT	STATE	SERVICE	VERSION	NOTE
3306/tcp	open	mysql	8.0.32	reused creds from portal config

DC01 Domain controller `10.10.20.5` Windows Server 2019 — Discovered

PORT	STATE	SERVICE	VERSION	NOTE
88/tcp	open	kerberos		
389/tcp	open	ldap		
445/tcp	open	smb		

WS-042 Workstation `10.10.20.42` — Not checked

No ports recorded.

Activity log

Changes recorded on the network map, plus lines written in the log, oldest first

MONDAY, OCTOBER 19, 2026

04:26 AM	Case started with 4 devices on the map	
04:36 AM	NOTE Started full TCP scan of 10.10.20.0/24	
04:46 AM	Intranet portal: ports 22/tcp (ssh), 80/tcp (http) added	Intranet portal
05:06 AM	Added Domain controller	DC01
05:07 AM	DC01: IP / host set to 10.10.20.5	DC01
05:36 AM	Intranet portal: status Discovered → Vulnerable	Intranet portal
06:01 AM	Linked Edge firewall → Intranet portal (Attack path)	Intranet portal
06:16 AM	Intranet portal: status Vulnerable → Compromised	Intranet portal
06:36 AM	Added Database	MySQL server
06:38 AM	Linked Intranet portal → MySQL server (Pivot / tunnel)	MySQL server
06:56 AM	MySQL server: status Not checked → Vulnerable	MySQL server
07:06 AM	Added Workstation	WS-042

My actions

What I did and the result, oldest first

#1 Recon Info Intranet portal Oct 19, 2026, 07:36 AM

Full TCP scan of the subnet

> 6 live hosts; web portal on .15 exposes 22 and 80

#2 Testing High Intranet portal Oct 19, 2026, 07:36 AM

Tested login form for SQL injection

> Boolean-based SQLi in username

#3 Findings Critical MySQL server Oct 19, 2026, 07:36 AM

Read app config through the injection

> DB credentials reused on MySQL

Notes & material

Oldest first

Oct 19, 2026, 05:46 AM

Login form returns a different error for valid usernames. Boolean-based SQLi confirmed in the username field.

Oct 19, 2026, 06:41 AM

Portal config file holds DB credentials (redacted here). Same password works on MySQL.